

Product Requirements Document (PRD)

FinPilot MCP – AI Investment Research Platform

Version: 1.0

Prepared For: NitroStack MCP Hackathon

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1. Product Overview

Product Name: FinPilot MCP

Tagline: Institutional-Grade Investment Research, Powered by Multi-Agent AI.

Vision:

FinPilot MCP is an AI-powered investment research platform that leverages the Model Context Protocol (MCP) to orchestrate multiple specialized AI agents. Instead of navigating multiple financial platforms, users can ask investment questions in natural language and receive comprehensive, evidence-backed investment reports.

2. Problem Statement

Investors rely on multiple disconnected platforms such as stock screeners, financial statement websites, annual reports, market news portals, and portfolio trackers. This fragmented workflow is time-consuming and often leads to incomplete analysis.

3. Solution

FinPilot MCP provides a conversational AI interface backed by specialized AI agents coordinated through MCP. Each agent performs a focused task such as market data retrieval, financial analysis, news summarization, valuation, or report generation. Their outputs are combined into a single investment report.

4. Objectives

- Simplify investment research
- Demonstrate MCP-based multi-agent orchestration
- Generate explainable investment reports
- Analyze companies and portfolios
- Reduce research time while improving decision quality

5. Core Features

- Company Research
 - Financial statements
 - Quarterly results
 - Annual reports
 - Financial ratios

- Historical price data
- Recent market news

- Investment Analysis

- Revenue growth
- Profitability
- Cash flow
- Debt analysis
- Valuation
- Risk assessment

- Portfolio Analysis

- Manual portfolio upload/input
- Diversification analysis
- Sector allocation
- Concentration risk
- Portfolio impact of new investments

Future Enhancement:

Direct brokerage integrations using MCP connectors.

- AI Investment Report

- Investment recommendation
- Business overview
- Financial summary
- Growth analysis
- Valuation summary
- Risk assessment
- Portfolio impact
- Confidence score

6. Multi-Agent Architecture

1. Orchestrator Agent
2. Market Data Agent
3. Financial Analysis Agent
4. News Intelligence Agent
5. Valuation & Risk Agent
6. Report Generator Agent

Workflow:

User → Orchestrator → Specialized Agents → Report Generator → Final Investment Report

7. Technology Stack

Frontend: Next.js + React

Backend: Python (FastAPI)

MCP Framework: NitroStack MCP SDK

AI Models: GPT-5.5 / Claude

Database: PostgreSQL

APIs: Financial Market APIs, Financial Statement APIs, News APIs

8. Future Scope

- Brokerage integrations via MCP
- Live portfolio synchronization
- Capital gains and tax intelligence
- Financial document OCR
- Earnings call analysis
- AI watchlists
- Real-time alerts

9. Success Criteria

- Investment report generated within 30 seconds
- Successful orchestration of multiple AI agents
- Evidence-backed recommendations
- Scalable MCP architecture for future integrations